

Gradient descent: This is a technique to estimate the parameters θ_0 and θ_1 of the hypothesis. The gradient descent equation for linear regression using a single variable is given below:

repeat until convergence: {
 $\theta_0 := \theta_0 - \alpha (1/m) \sum (h_0(x_i) - y_i)$, for $i=1$ to m
 $\theta_1 := \theta_1 - \alpha (1/m) \sum ((h_0(x_i) - y_i) x_i)$, for $i=1$ to m
}

...where, m is the size of the training set and x_i , y_i are values of the given training set (data).

Regularisation: This keeps all the features but reduces the weightage.

$\min_{\theta_0, \theta_1} \frac{1}{2m} [\sum (h_0(x^{(i)}) - y^{(i)})^2 + \lambda \sum (\theta_j)^2]$, for $i=1$ to m and $j=1$ to n

λ , or lambda, is the regularisation parameter. It determines how much the cost of the θ parameters is inflated in order to reduce the weightage of the feature in the hypothesis function.

Evaluating a hypothesis

There is a misconception that the algorithm that produces low errors is also accurate. This may not always be the case, and the main reason for this could be overfitting. Overfitting (high variance) is caused by a hypothesis function that fits the available data well but does not generalise to predict new data. So it is very important to understand whether the algorithm is overfitting or underfitting (high bias), when the hypothesis function (h) maps poorly to the trend of the data. The following section provides some general guidelines to evaluate various aspects of the hypothesis/algorithm.

Model selection: Model selection is the process by which we choose the hypothesis with the least errors (more generalised), set parameters and identify any pre-processing requirement to predict the best results. However, it has been observed that errors will always be low on the data used for training the model compared to any other data. Hence, experts recommend another approach using a cross-validation set as detailed below.

- Split the dataset into three sets—a training set, a test set and a cross-validation set. The possible breakdown could be 60-20-20.
- Optimise the parameters in θ using the training set.
- Find the polynomial degree d with the least error using the cross-validation set.
- Estimate the generalisation error using the test set.

In this way, the degree of the polynomial (d) has not been trained using the test set and, hence, the generalisation error can be minimised.

Diagnosis of bias vs variance: The study of relationships between the cost function and the degree of the polynomial (d) can help us determine whether underfitting or overfitting is the problem with our hypothesis. Here are some trends to look

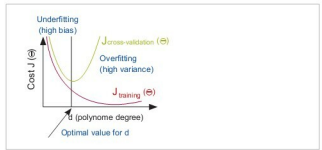


Figure 1: Diagnosis of bias + variance

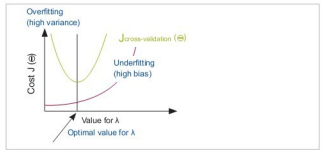


Figure 2: Regularisation and bias + variance

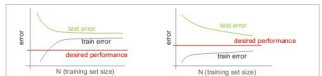


Figure 3: Learning curves

for during the diagnosis of bias and variance.

- Errors on training data decrease when we increase the degree of polynomial (d).
- Cross-validation errors decrease on increasing (d) up to a certain point, and then start to increase again. Figure 1 summarises these trends.

Underfitting happens when (d) is small, and errors on both training data and cross-validation data are high. Overfitting happens when (d) is large, and training data errors are low while cross-validation errors are comparatively higher.

In case of regularisation, we can use the regularisation parameter λ to determine high bias/variance. The rule of thumb to use is:

- Large λ : High bias (underfitting)
- Intermediate λ : Just right
- Small λ : High variance (overfitting)

The relationship of λ to the training and cross-validation errors is shown in Figure 2.

Underfitting happens when λ is large, and errors on both training and cross-validation data are high.

Overfitting happens when λ is small, and errors on training data are low while cross-validation errors are high.

Learning curves: Learning curves can help in choosing the best value for (d) or λ to plot them. They can help us understand the relationship between errors and the test data size in situations of high bias/variance.